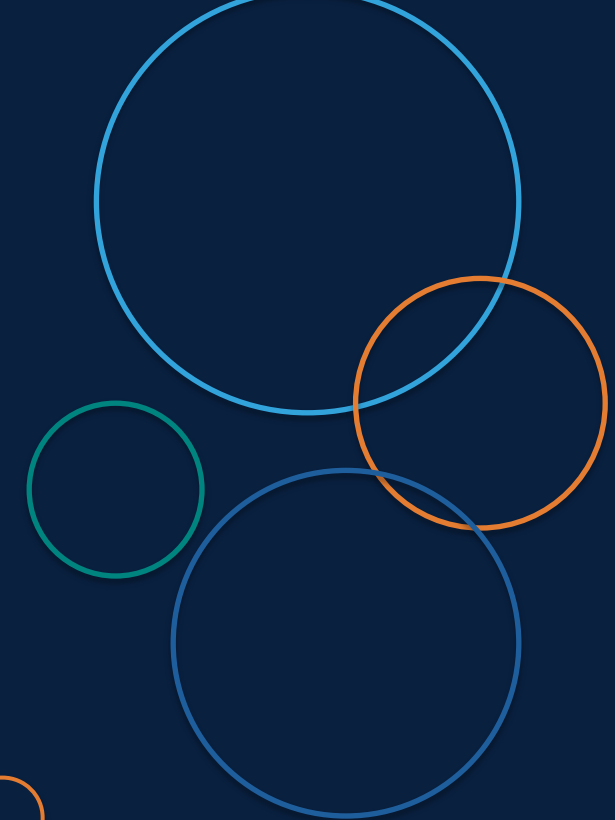


Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities



A data-driven overview of emissions from the Industrial Revolution to 2024

1750–2024

Scope of historical emissions
trajectory

429 ppm

Atmospheric CO₂ at Mauna Loa,
Feb. 2026

35+ Bt

Annual fossil fuel and industry
CO₂ today

Audience: policy analysts and climate negotiators | Sources: NASA; Our World in Data / Global Carbon Budget 2025

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

Latest Mauna Loa measurement (NOAA, Feb. 2026)

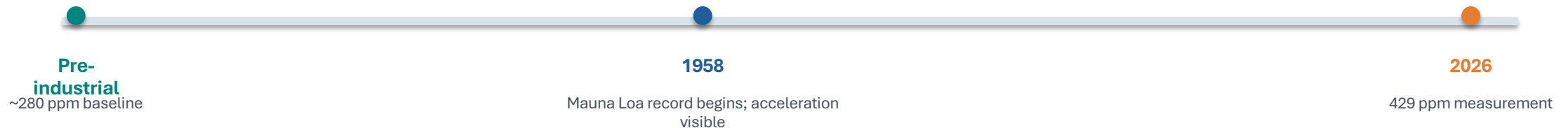
~280 ppm

Approximate pre-industrial baseline

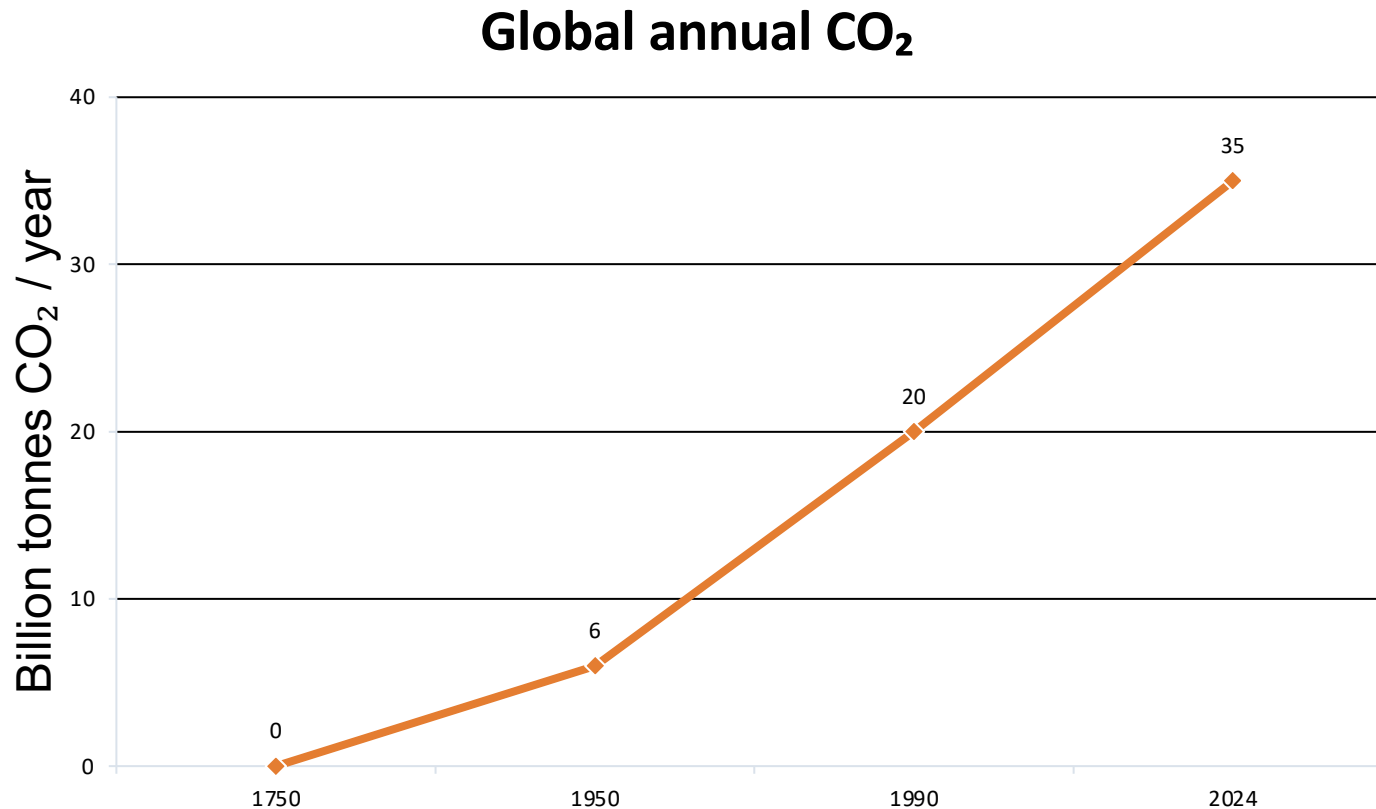
>50%

Increase above pre-industrial levels

- Ground-based and satellite measurements confirm a sharp acceleration linked primarily to coal, oil, and natural gas combustion.
- The same magnitude of natural increase after the last ice age took thousands of years; today's rise occurred in roughly two centuries.



Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes



Milestone line chart

- Emissions were negligible before the Industrial Revolution.
- Growth remained slow until the mid-20th century.
- By 1990, emissions had nearly quadrupled from 1950.
- Today, annual emissions exceed 35 Bt and have not yet peaked.

Reference: figure_2.jpg in outline; trajectory recreated from source brief milestones because asset was not provided.

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

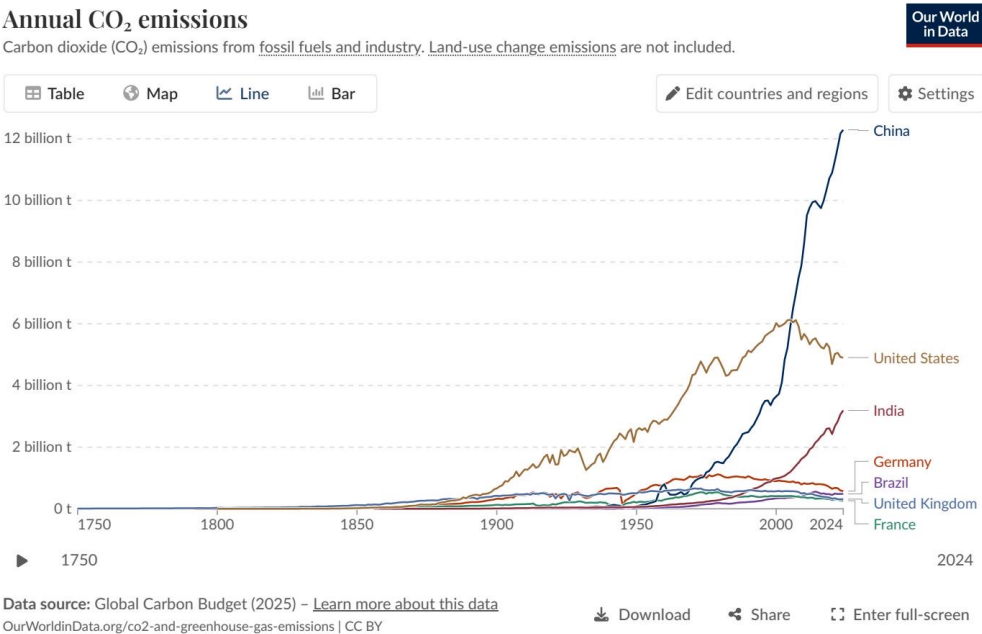
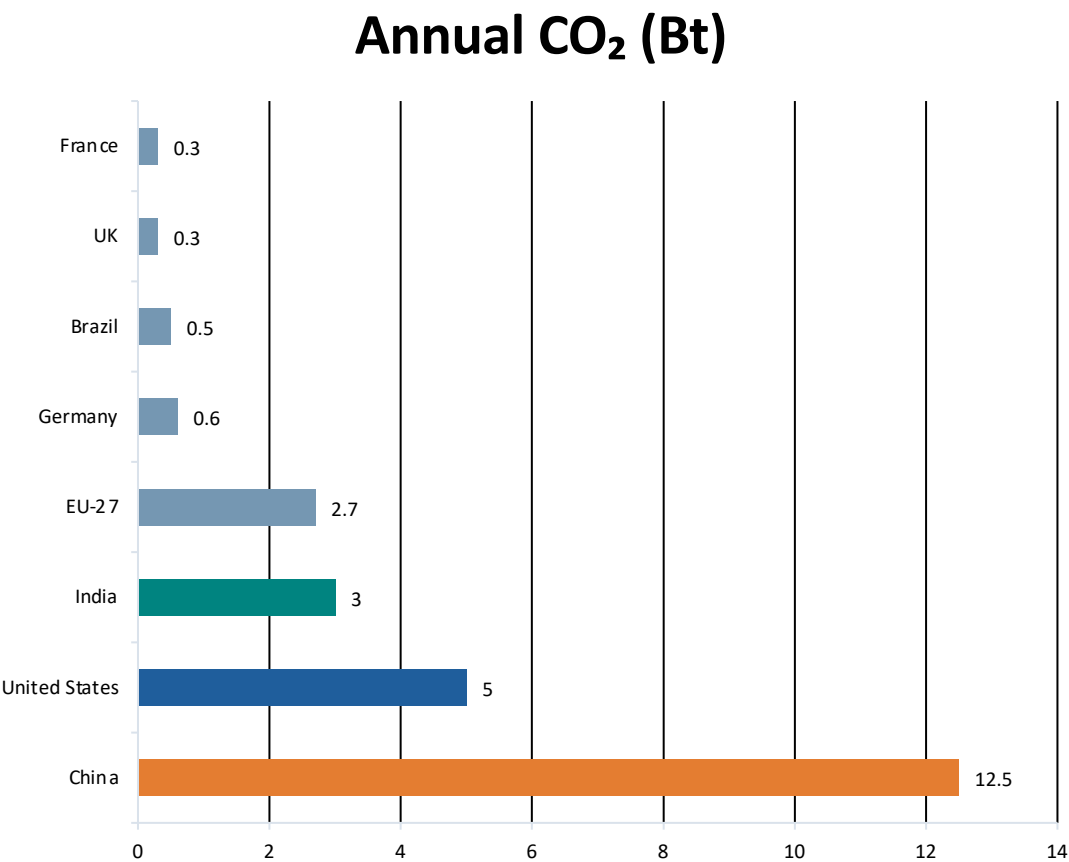
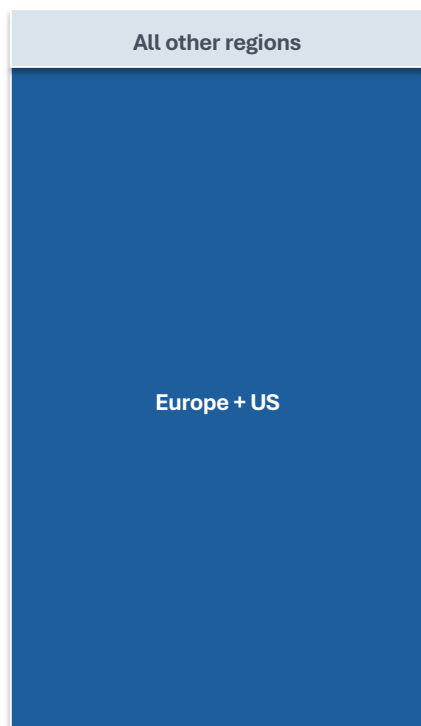


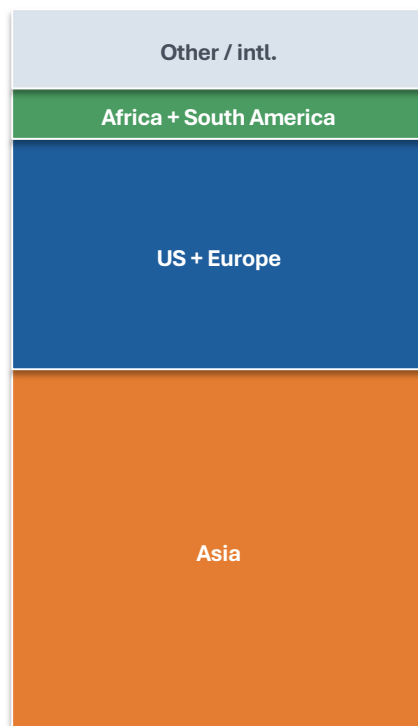
Figure reference: figure_3.jpg — Annual CO₂ emissions by country

- China produces over one-quarter of global emissions (~27%).
- The US peaked near 6 Bt around 2005 and declined to ~5 Bt.
- India has risen rapidly to approximately 3 Bt.

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third



1900



2024

Side-by-side comparison visual: exact shares are directional categories from the brief (>90%, ~50%, <33%, 3–4%).

>90%

Europe + US share in 1900

~50%

Asia share today

<1/3

US + Europe today

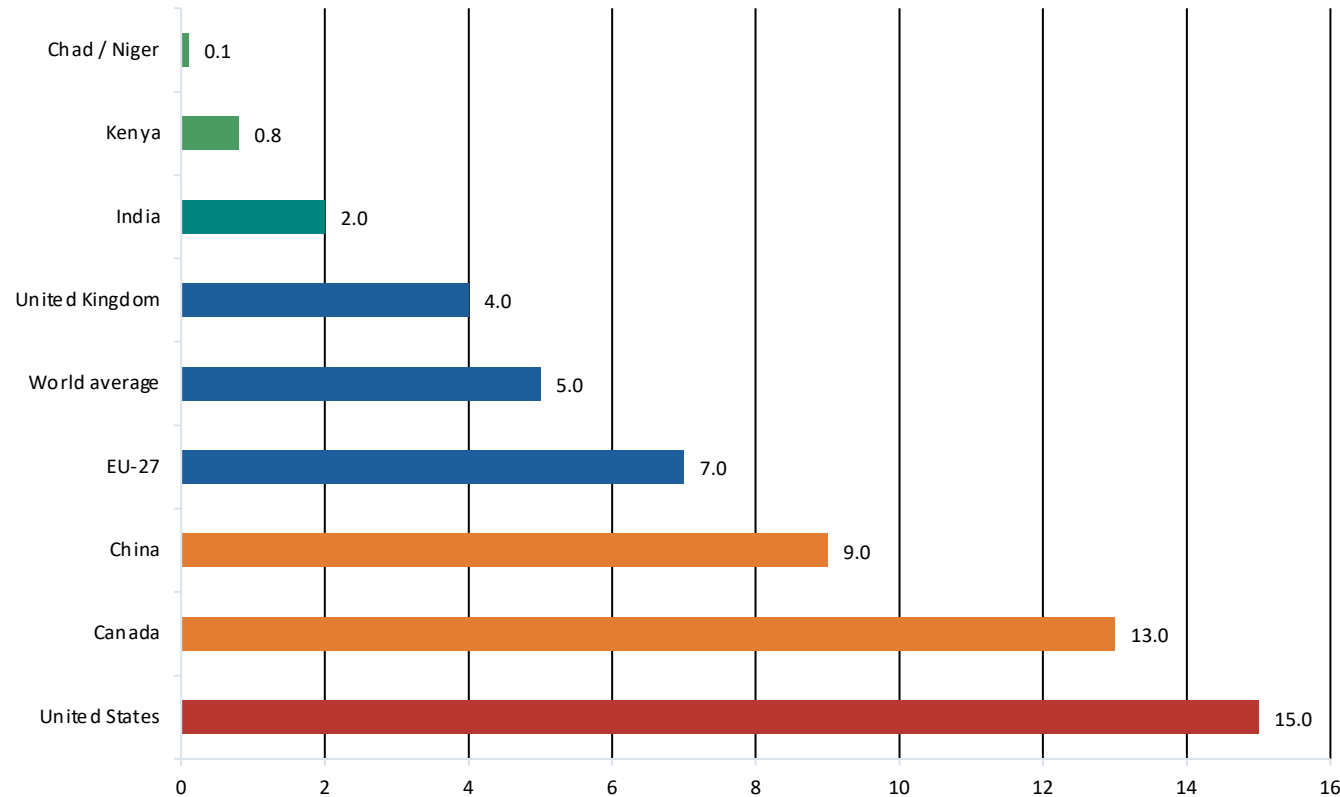
3–4%

Africa and South America each

- By 1950, Europe and the US still represented over 85% of global emissions.
- The geographic center of annual emissions has shifted toward Asia, which also holds nearly 60% of world population.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

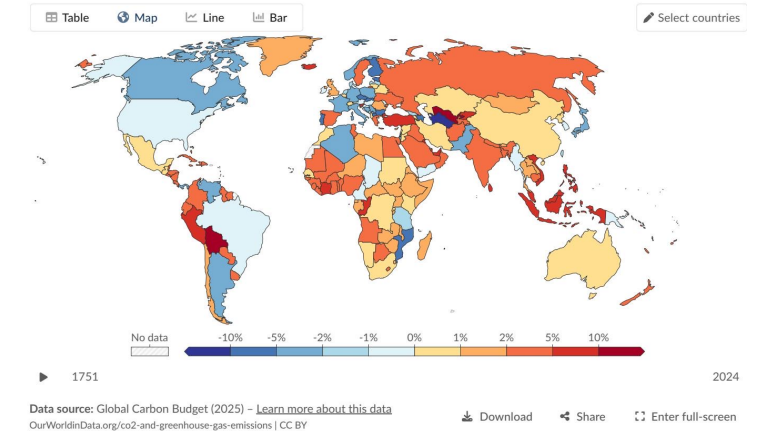
t CO₂/person/year



Kenya is shown as <1 t/person/year; chart uses a sub-1 visual placeholder to preserve ordering without implying exactness.

Annual percentage change in CO₂ emissions, 2024

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.



150×

Approximate gap
from US/Australia
to Chad/Niger

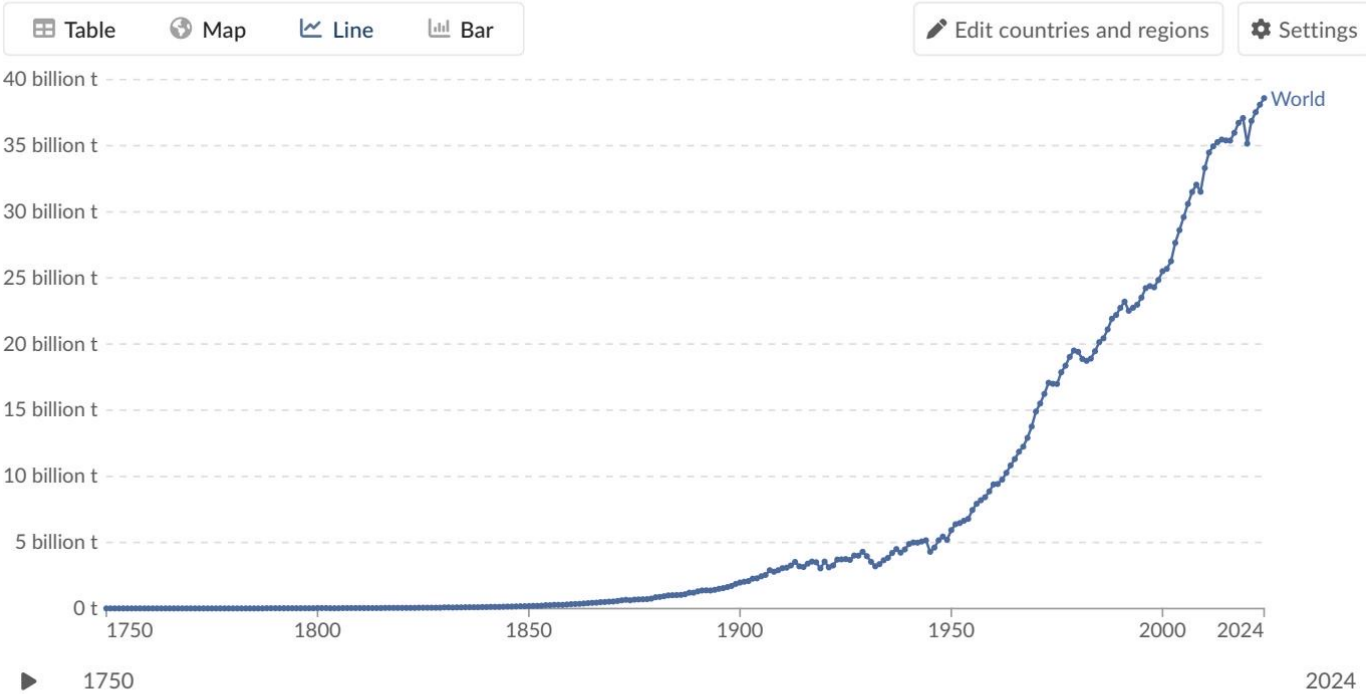
The average American or Australian emits in under two days what the average person in Chad or Niger emits in an entire year.

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data



Data source: Global Carbon Budget (2025) – [Learn more about this data](#)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

Download Share Enter full-screen

Figure reference: figure_4.jpg — Annual percentage change in CO₂ emissions, 2024

–1% to –5%
US and much of Europe recorded declines

+2% to +10%
Russia, parts of Africa, and SE Asia increased

- Blue areas show negative annual change across the United States and much of Europe.
- Orange and red areas show increases in Russia, Southeast Asia, and several African nations.
- Some South American countries show very large percentage increases (>10%), while others decline.

Interpretation: annual percentage change captures near-term direction, not cumulative responsibility.

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can produce very different per capita emissions when power and industrial systems rely on different energy sources.

Country / group	Energy-source pattern	Per capita CO ₂	Policy implication
France	Higher nuclear and renewable shares	~4 t/person	Low-carbon supply lowers per capita emissions
United Kingdom	Higher nuclear and renewable shares	~4 t/person	Declining emissions with lower per capita outcome
Germany	Higher fossil fuel share than France/UK	Higher than France/UK; not quantified in brief	Energy mix keeps per capita emissions higher
Netherlands	Higher fossil fuel share than France/UK	Higher than France/UK; not quantified in brief	Energy mix keeps per capita emissions higher



Note: the brief provides exact per capita values for France and the UK (~4 t/person), and qualitative contrast for Germany/Netherlands as higher due to fossil-fuel share.

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

- 1 Global emissions exceed 35 Bt/year and have not yet peaked.
- 2 China alone emits >25% of the global total (~12.5 Bt), more than double the US (~5 Bt).
- 3 Annual emissions geography has shifted from Europe + US dominance to Asia at roughly half today.
- 4 Per capita inequality remains extreme: ~15 t/person in the US versus ~0.1 t in Chad/Niger.
- 5 Energy mix and policy choices shape outcomes; prosperity alone does not determine per capita CO₂.

Negotiation lens: annual totals, per capita equity, historical shifts, and energy-system choices all describe different facets of responsibility.